***Brock Biology of Microorganisms, 15e* (Madigan et al.)**
Chapter 8   Viruses and Their Replication

8.1   Multiple Choice Questions

1) Viral replication is
A) independent of the host cell's DNA but dependent on the host cell's enzymes and metabolism.
B) independent of both the host cell's DNA and the host cell's enzymes and metabolism.
C) dependent on the host cell's DNA and RNA.
D) dependent on the host cell's DNA, RNA, enzymes, and metabolism.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.1

2) Viral replication occurs
A) intracellularly.
B) extracellularly.
C) both intracellularly and extracellularly.
D) either intracellularly or extracellularly, depending on the virus involved.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.1

3) Viral size is generally measured in
A) micrometers.
B) picometers.
C) nanometers.
D) centimeters.
Answer: C
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.2

4) Enveloped viral membranes are generally ________ with associated virus-specific ________.
A) lipid bilayers / phospholipids
B) protein bilayers / lipids
C) lipid bilayers / proteins
D) glycolipid bilayers / phospholipids
Answer: C
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.2

5) Which statement is TRUE?
A) All viruses contain their own nucleic acid polymerases.
B) RNA viruses contain their own nucleic acid polymerases.
C) Viruses do not contain their own nucleic acid polymerases.
D) The origins of the nucleic acid polymerases used by viruses are eukaryotic.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.2

6) Reverse transcriptase is a(n)
A) RNA-dependent DNA polymerase.
B) DNA-dependent DNA polymerase.
C) RNA-dependent RNA polymerase.
D) DNA-dependent RNA polymerase.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.2

7) Viruses infecting ________ are typically the easiest to grow in the laboratory.
A) plants
B) animals
C) fungi
D) prokaryotes
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.4

8) all viral particles
A) are metabolically inert.
B) are smaller than bacterial cells.
C) contain an envelope to prevent its degradation outside of a host.
D) exhibit cell lysis under a particular condition.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.1

9) Cellular receptors may be composed of
A) proteins.
B) carbohydrates.
C) lipids.
D) combinations of proteins, carbohydrates, and/or lipids.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.5

10) Restriction is
A) the viral process whereby a host′s DNA ceases normal functioning.
B) the viral process whereby the virus prevents other viruses from entering the cell.
C) a general host mechanism to prevent the invasion of foreign nucleic acid.
D) a general host mechanism to prevent virus particles from further infective action.
Answer: C
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.5

11) All of the following can act as receptors for various bacteriophages EXCEPT
A) iron transport proteins.
B) flagella.
C) Sugar transporters.
D) cilia.
Answer: D
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.5

12) The T4 bacteriophage could not infect *Staphylococcus aureus* because this bacterium does NOT possess a
A) lipopolysaccharide outer membrane.
B) teichoic acid outer membrane.
C) pili.
D) flagella.
Answer: A
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.5

13) Bacteriophages′ genomes are typically composed of
A) single-stranded RNA.
B) single-stranded DNA.
C) double-stranded RNA.
D) double-stranded DNA.
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.5

14) A virus that kills its host is said to be
A) lytic or virulent.
B) temperate.
C) lysogenic.
D) virulent or lysogenic, but not temperate.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.7

15) Which of the following enzymes would you expect to find in the virion of a retrovirus, but NOT in a bacteriophage?
A) lysozyme
B) methylase
C) restriction enzymes
D) reverse transcriptase
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.8

16) The packaging mechanism of T4 DNA involves cutting of DNA from
A) linear genetic elements.
B) circular genetic elements.
C) DNA concatemers.
D) its host cells.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.6

17) A prophage replicates
A) along with its host while the lytic genes are expressed.
B) along with its host while the lytic genes are not expressed.
C) independently of its host while the lytic genes are expressed.
D) independently of its host while the lytic genes are not expressed.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.6

18) The life cycle of lambda phage is controlled by accumulation of repressor proteins. For lysogeny to occur
A) cI protein predominates.
B) Cro protein predominates.
C) cII protein is repressed.
D) cI protein is repressed.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.7

19) Which of the following statements is FALSE?
A) Lambda is a temperate phage that infects *Escherichia coli*.
B) Lambda is a linear double-stranded DNA phage.
C) Lambda is replicated by the rolling circle mechanism.
D) Lambda always circularizes upon entering the host cell.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.7

20) Retroviruses are medically important viruses because
A) they include the poliovirus.
B) they include the influenza virus.
C) they include all human pathogenic viruses.
D) they include some viruses that cause cancer and HIV.
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.8

21) The genome of retroviruses contains genes to make all of the following EXCEPT
A) structural proteins.
B) repressor proteins.
C) integrase.
D) proteases.
Answer: B
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.8

22) Rolling circle replication of the lambda genome differs from replication of a bacterial chromosome in that
A) bidirectional replication forks are not formed.
B) only a single strand of the genome is copied.
C) no concatamers are formed.
D) only a single strand of the genome is copied and no concatamers are formed.
Answer: A
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.7

23) As a consequence of infection by a temperate bacteriophage such as lambda, the host cell
A) lyses as a result of bacteriophage release.
B) never lyses but continues to divide and replicate both itself and the prophage.
C) divides faster at moderate temperatures.
D) may lyse or may continue to divide and replicate both itself and the prophage.
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.7

24) When packaged in the virion, the complete complex of nucleic acid and protein is known as the virus
A) capsid.
B) concatemer.
C) nucleocapsid.
D) envelope.
Answer: C
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.1

25) Which of the following are the hosts for most enveloped viruses?
A) *Bacteria*
B) animals
C) *Archaea*
D) fungi
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.1

26) Prophage formation occurs after lambda infection if
A) the cI gene is transcribed and cI protein accumulates.
B) cII expression occurs.
C) Cro accumulates in the cell.
D) both Cro and cI are equally activated.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.7

27) The HIV genome consists of
A) a single ds RNA molecule.
B) two identical ssRNA molecules.
C) two identical ssDNA molecules.
D) a single DNA molecule.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.8

28) When a virus enters a host cell in which it can replicate, the process is called a(n)
A) insertion.
B) infection.
C) prophage.
D) excision.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.1

29) The term ″phage″ is generally reserved for the viruses that infect
A) animals.
B) plants.
C) bacteria.
D) multiple species.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.1

30) The size and shape of viral particles is largely governed by the size and packaging of the viral
A) envelope.
B) enzymes.
C) prophage.
D) genome.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.2

31) You are attempting to mutate lambda to affect whether lysis or lysogeny occurs after infection. Which mutation would INCREASE the chances of LYSOGENY over lysis?
A) deletion or inactivation of the *cI* gene
B) deletion or inactivation of the *cro* gene
C) overexpression of the *cro* gene
D) deletion of both the *cro* and *cI* genes
Answer: B
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.7

32) Regarding the viral membrane of an enveloped virus, the lipids are derived from the ________, and the proteins are encoded by ________.
A) host′s cell membrane / viral genes
B) virion / viral genes
C) host′s cell membrane / host′s genes
D) virion / host′s genes
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.1

33) Some bacteriophage possess an enzyme similar to ________, which makes a small hole in the bacterial cell wall, allowing the viral nucleic acid to enter.
A) peptidoglycanase
B) infectase
C) lysozyme
D) nuclease
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.2

34) The use of ________ is the easiest and most effective way of studying many animal and plant viruses.
A) bacterial cultures
B) tissue or cell culture
C) live hosts
D) prophages
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.4

35) When a solution composed of bacteria and infectious virions are mixed and spread on an agar plate, ________ form where viruses lyse the host cells.
A) insertion sequences
B) plaques
C) prophages
D) colonies
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.4

36) T4 genes are transcribed by host RNA polymerase, yet this transcription is carefully controlled so that groups of T4 genes are transcribed in a specific order after infection. How is this accomplished?
A) Early T4 genes encode for proteolytic enzymes that destroy the host RNA polymerase. Subsequently a viral polymerase is created that transcribes the middle and late genes in the correct order.
B) Early and middle T4 genes encode for RNA polymerase-modifying proteins so that only phage promoters are recognized.
C) Each group of T4 genes has a different promoter that indicates the order in which they should be transcribed based on the affinity of the promoter for the host RNA polymerase.
D) Rolling circle replication of the viral genome ensures that the genes are available for transcription in the correct order.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.6

37) The T4 phage protects its DNA from host restriction endonucleases by
A) glucosylating cytosine bases in the T4 genome to prevent DNA cleavage.
B) methylating all four bases (A, T, C, G) in the T4 genome to prevent DNA cleavage.
C) integrating the viral genome into the host genome where it will not be degraded.
D) circularizing the viral genome so that it will not be degraded.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.5

38) In *E. coli*, the adenine in the sequence GATC is methylated by the Dam enzyme. In the same cells, a restriction endonuclease recognizes and cleaves dsDNA with the sequence GATC on either strand. Why does *E. coli* have these two enzymes?
A) The enzymes cut the *E. coli* genome into pieces that bind to viral particles and inhibit viral replication.
B) The enzymes increase the rate of mutation and genome rearrangement, thus increasing the likelihood that *E. coli* cells will mutate and become resistant to viral infection.
C) The enzymes encourage lysogeny because the cleavage sites are recognized by viral integrases.
D) The enzymes protect *E. coli* from infection by preferentially degrading viral or other exogenous DNA that is not methylated.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.5

39) The majority of many important human viral diseases are caused by
A) ssRNA viruses.
B) ds RNA viruses.
C) ss DNA viruses.
D) ds DNA viruses.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.8

40) Viral proteins are categorized as early, middle, and late. Early proteins typically are necessary for
A) production of viral mRNA.
B) packaging of DNA into the nucleocapsid.
C) copying the viral genome.
D) production of viral mRNA and copying the viral genome.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.6

41) A cell that allows the complete replication cycle of a virus to take place is said to be a
A) permissive host.
B) viral cell.
C) dead cell.
D) lytic cell.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.3

42) In a natural population of diverse slow-growing prokaryotic cells, what type of viruses would you expect to be most common?
A) lytic bacteriophages
B) enveloped viruses
C) icosahedral viruses
D) temperate bacteriophages
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.7

43) The general steps of the viral lifecycle are similar in most viruses. One major exception, however, is entry into the host cell. How does this step differ between an animal cell and *E coli*?
A) The entire virion is taken into an animal cell, but only the viral genome enters *E coli*.
B) The entire virion enters *E coli*, but only the nucleic acid is taken up by animal cells.
C) The virion fuses to the bacterial cell membrane of *E coli*, while the genome is injected into an animal cell.
D) The virion randomly attaches to *E coli*, but viral binding to a specific receptor is required for entry into animal cells.
Answer: A
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.8

44) What would be the consequence of deleting the late T4 genes?
A) The T4 genome would not be copied.
B) T4 mRNA would not be produced.
C) T4 capsid proteins would not be made.
D) ATP would not be produced and the T4 genome would not be packaged into the capsid.
Answer: C
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.6

45) Prokaryotic restriction endonucleases are effective at destroying a virus whose genome consists of
A) ds DNA.
B) ssDNA.
C) dsRNA.
D) ssRNA.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.5

46) You isolate a bacteriophage that can replicate in *E. coli*. Through chemical analyses you determine that the only nucleic acid present is RNA. You isolate the RNA and put it in a test tube with all of the enzymes, amino acids, and RNAs necessary for translation. The RNA is translated directly, without being copied into a complementary strand first, and new infectious virions are made and released. What does this tell you about the bacteriophage?
A) The viral genome is ssRNA of the plus sense.
B) The viral genome is ssRNA of the minus sense.
C) The bacteriophage is a retrovirus.
D) The bacteriophage is probably a new strain and should be studied further.
Answer: A
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.5

47) For bacteriophages and animal viruses ________ is the step in the viral life cycle that determines host cell specificity.
A) attachment
B) penetration
C) synthesis
D) assembly
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.5

48) What are the possible consequences of viral infection of an animal cell?
A) rapid lysis or latent infections
B) lysogeny followed by eventual lysis
C) lysis or lysogeny
D) Outcomes vary from rapid lysis to persistent infections, latent infections, or cancer.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.8

49) For a virus to cause a latent infection, it must possess
A) ssDNA.
B) ds DNA.
C) dsRNA.
D) Any type of viral genome can lead to a latent infection.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.8

50) The concentration of infectious plaque forming units (pfu) per volume of fluid is known as the
A) infectivity.
B) virulence.
C) titer.
D) fluid infectivity.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.4

51) The growth of viruses in a culture is described as a one-step growth curve, because
A) virion numbers show no increase during intracellular replication and can only be counted after the virions burst from the host cell.
B) there is only one step in the viral life cycle which leads to only one replicative cycle in a culture.
C) assembly and release actually occur in one step.
D) the eclipse phase prevents the plating and enumeration of virions although new virions are produced at a steady rate during the eclipse phase.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.3

8.2 True/False Questions

1) Viruses can confer additional properties on their host cells, which can in turn be inherited.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.8

2) In a population of T4 virus, each virion has the same set of genes, but they are arranged in a different order.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.6

3) Accumulation of the cI protein results in integration of the lambda genome and prophage formation.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.7

4) Virus assembly requires host cell mediated reactions.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.6

5) Budding of virions from an infected host results in enveloped virus.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.8

6) Rod-shaped viruses have icosahedral symmetry while spherical viruses have helical symmetry.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.2

7) The genome of a temperate phage can replicate along with the host genome during lysogeny.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.7

8) Penetration requires that the entire virus is inserted within the host.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.5

9) RNA viruses encode host restriction systems designed to destroy host DNA.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.8

10) The genome of a retrovirus consists of dsRNA.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.8

11) Although T4 encodes over 250 proteins, it does not encode its own RNA polymerase.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.6

12) Temperate viruses can enter into either a lytic or lysogenic cycle.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.7

13) Lysogeny is unique to bacteriophages; similar relationships have not been found among the animal viruses.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.8

14) A naked virus lacks a capsid.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.1

15) The reason dogs do not get measles is because their cells lack the correct receptor sites for that virus.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.8

16) A lytic infection results in death of the host cell.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.3

17) An RNA genome itself serves as mRNA in negative-stranded RNA viruses.
Answer: FALSE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.2

18) The latent phase in the viral growth curve and the lag phase of the bacterial growth curve are equivalent and represent the time it takes for the virus or bacterium to adapt to the culture conditions and begin growing.
Answer: FALSE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.3

19) In prokaryotes, DNA viruses replicate their genomes in the nucleus while RNA viruses are replicated in the cytoplasm.
Answer: FALSE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.8

8.3   Essay Questions

1) Why have the majority of viruses evolved to bind to a host surface receptor that serves an essential function in the host cell?
Answer: Host surface receptors that are essential for the host will always be expressed and present on the host. This means a virus that binds to this receptor will always have a way of gaining entry into the cell to replicate.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.5

2) Compare and contrast the structure, life cycle, and host cell type of naked and enveloped viruses.
Answer: A naked virus lacks an envelope and is called a nucleocapsid, because it contains only a nucleotide-based genome and capsid proteins. A virus containing a nucleocapsid enveloped within an additional lipid layer is called an enveloped virus. Most enveloped viruses are animal viruses, while naked viruses infect animal, plant, and prokaryotic cells. The penetration phase of the viral life cycle differs between enveloped and naked viruses. The entire enveloped viral particle typically enters an animal host cell during penetration, while many naked viruses only inject their genome during penetration. In addition, release of enveloped viral particles usually occurs through budding so that the host cell membrane drapes around the nucleocapsid. Naked viruses are more often released through cell lysis, especially in cell types with cell walls.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.8

3) Explain the potential advantages of lysogeny versus lysis for a temperate virus.
Answer: Answers could vary, but one idea is that lysogeny allows the virus to increase in number within the host without killing it. This would be advantageous if there are few host cells present (i.e. the host population is small). Lytic viruses or lytic cycles expel viruses that then must attach to a new host to proliferate. If the viral host population is small, a lytic cycle might kill or wipe out the entire host population and prevent further viral replication.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.7

4) How does plating efficiency affect the number of plaque-forming units? How is plating efficiency calculated?
Answer: The issue of plating efficiency is that not every virus will be infective or able to lyse the host in the medium. Consequently, the number of plaques observed is expected to be less than the actual number of viral particles in the culture. Determination of plating efficiency is therefore necessary when counting plaque-forming units per volume plated to calculate the actual viral concentration (or titer). Plating efficiency is determined by comparing the number of plaques observed to the number of viral particles observed with an electron microscope in the same culture.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 8.4

5) Relate the structure of bacteriophages and animal viruses to the structure of their respective host cells and the steps of the viral life cycle.
Answer: The T4 phage has a very complex structure, containing a head, collar, tail, tail pins, an endplate, and tail fibers. The tail fibers bind to specific receptors on the surface of a bacterial cell (outside the peptidoglycan cell wall). When the tail fibers retract, the tail pins bind to the host so that the tail itself interacts with the cell wall. A viral enzyme then forms a small pore in the host cell wall. The tail′s covering then contracts and inserts the virus into the host′s cytoplasm. Animal viruses are usually much simpler in structure than bacteriophages because animal host cells lack a cell wall. Penetration is therefore somewhat easier and is usually accomplished by endocytosis or merging of the viral envelope with the host cell membrane. The complicated fibers and contractile tail are not needed to penetrate the animal cell. One last difference is that the entire virion often enters the animal cell and not just the viral genome.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.8

6) Why was the discovery of retroviruses important to the field of molecular biology?
Answer: The discovery of retroviruses was important to the field of molecular biology, because it changed the central dogma of molecular biology that states that information in a cell flows from DNA→RNA→protein. Retroviruses have an RNA genome that is converted to DNA, which reverses the flow of information. Although other viruses have RNA genomes, the RNA is not converted to DNA in other RNA viruses and thus the central dogma was not changed. The discovery of retroviruses also revealed a new kind of nucleic acid polymerase—an RNA-dependent DNA polymerase. This enzyme has subsequently been used extensively in genetic engineering and molecular biology research to facilitate the study of RNA molecules by converting them to DNA.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.8

7) T4 and other bacteriophages commonly use a method of DNA replication and packaging called ″headful packaging.″ Explain how viral genomes are replicated and packaged in this process and hypothesize how headful packaging might affect evolution of BOTH the bacteriophages and their prokaryotic hosts.
Answer: A linear concatemer is first formed after recombination of several copies of the viral genome. An endonuclease then cuts out particular regions of the concatemer to form individual linear replicated genomes. Although the genomes will all contain the same genes, this circular permutation process results in genes being in a different sequence. In addition, the concatemer is not cut at specific sequences but is instead cut into linear segments of DNA just long enough to fill the phage head. The phage head typically holds slightly more than a genome length of DNA, thus the mechanism generates terminal repeats at each end of the DNA genome. Headful packaging may allow for the mixing and recombination of phage genes because not every page head has an identical genome. This might increase the rate of evolution of the phage. In addition, this mechanism of packaging is not sequence specific, thus host DNA could be packaged into the phage head instead of viral DNA. When the phage infects another bacterial cell, it would inject new bacterial genes into the host cell. Without a full complement of phage genes, the phage would not replicate within the new host cell, but the new bacterial genes could add to or change the genetic material of the host. Therefore, headful packaging could increase the evolution of the prokaryotic host as well.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.6

8) Describe the establishment and maintenance of the lysogenic state in the lambda phage.
Answer: Lambda phage, once circularized by matching its cohesive ends, integrates into an *E. coli* host genome by targeting the *attλ* gene in the host genome. The *att* gene in lambda′s genome then base pairs with the complementary sequences found in the *attλ* gene, and the lambda genome is inserted in the presence of a viral integrase. After the lambda genome is established in the host′s genome, it is maintained along with the *E. coli* genome through replication. Lysis is triggered by DNA damage or other cellular stress and involves a viral excision enzyme.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 8.7

9) Influenza is an acute human viral disease that causes brief cellular damage followed by healing and complete clearing of the virus from the body. Hepatitis C is a chronic viral disease that causes slow destruction of liver tissue and persistent virions that are not completely cleared. Which of these diseases is more likely to be caused by lytic virus and which is caused by a non-lytic virus?
Answer: Persistent viral infections occur when the host cell remains alive and continually produces more viruses. These conditions are especially common in animal cells with a temperate virus that can form a prophage within the host genome, thus hepatitis C is more likely caused by a lysogenic or temperate virus. Based on the short duration of symptoms with influenza and the disappearance of the virus after symptoms stop, influenza is more likely to be caused by a lytic virus.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.8

10) Explain why viruses are not considered to be living entities.
Answer: A virus consists of genetic material surrounded by a protein coat (capsid) and in some animal viruses an outer envelope. Considered obligate intracellular parasites, viruses can only replicate inside a living host cell. They must rely on the host cell for energy, metabolic intermediates and protein synthesis. While all cells possess ds DNA as the genetic material, some viruses utilize ssDNA, dsRNA, or ssRNA.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.1

11) Explain why some viruses contain enzymes within the virion and others do not. Use specific examples to support your answer.
Answer: Viruses do not carry out metabolic processes and thus are metabolically inert. However, some viruses carry enzymes in their virion that play important roles in infection. In the example of retroviruses, the reverse transcriptase enzyme, which is critical for its replication is found within the virion itself. Bacteriophages contain lysozyme in the capsid to penetrate the peptidoglycan cell wall. However, many other viruses do not carry enzymes simply because they use the host's cellular machinery to translate enzymes that they require.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.2

12) Viruses variously depend on their host cell for parts of the viral replication and maturation process. Many viruses do however encode for some enzymes that facilitate certain steps or virus-specific processes for viral replication and maturation. Which cellular processes and enzymes do all viruses lack and must therefore ALWAYS be supplied by the host cell?
Answer: In general ALL currently described viruses lack (i) metabolism and (ii) translational machinery. To elaborate, all viruses lack the ability to make ATP, therefore the energy for replication and maturation of viral particles must be supplied by the host cell. Put another way, all viruses lack metabolism or metabolic enzymes capable of generating ATP or any other form of energy for anabolism. Second there are no currently described viruses that have their own ribosomal proteins, rRNA, or tRNA. All portions of the translational apparatus for synthesizing proteins are supplied by the host cell. Other enzymes, such as RNA polymerase, DNA polymerase, etc. MAY be supplied by the host cell or may be encoded for in the viral genome and may even be present in the viral capsid.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.10

13) Propose a unique target for a drug that would inhibit replication of retroviruses in humans.
Answer: Students could choose to discuss protease, integrase, or reverse transcriptase enzymes, all of which are found in the nucleocapsid of retroviruses and are unique enzymes that have no eukaryotic equivalents. Answers should explain why their target does not have an equivalent in the human host cell. Students could also discuss cell surface receptor proteins, although these proteins were not discussed in detail in this chapter.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.8

14) Predict the properties of viral populations in the human colon given that the colon contains a very high concentration of rapidly dividing bacteria. Predict as many viral properties as you can with this information.
Answer: The large intestine contains bacteria belonging to over 30 identified genera. Consequently, viral populations in the human colon are most likely dominated by head and tail lytic bacteriophages. Rapidly dividing bacterial populations tend to favor the survival and replication of more lytic phages whereas slowly dividing small populations favor lysogenic phages.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.5

15) Although bacteria lack an immune system, they do possess several weapons against viral attack. Explain the mechanisms utilized by bacteria to prevent viral infection.
Answer: (1) When bacteriophage interrupt the host cell's transcription and translation, a toxin-antitoxin system may be activated that limits the phage's replication. (2) Clustered regularly interspaced short palindromic repeats (CRISPR) are segments of prokaryotic DNA containing short repetitions of base sequences. Each repetition is followed by short segments of "spacer DNA" from previous exposures to a bacteriophage virus orplasmid. CRISPR associated proteins (Cas) use the CRISPR spacers to recognize and cut these exogenous genetic elements out. This confers resistance to foreign genetic elements such as those present within phages, and provides a form of acquired immunity. (3) Additionally, *Bacteria* can destroy viral ds DNA through the activity of restriction endonucleases. These enzymes cleave foreign DNA at specific sites and consequently prevent invasion.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 8.5